

Risk-Aware Distributional Reinforcement Learning for 5G-Advanced Handover Decisions

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Abstract—Handover (HO) decisions in 5G-Advanced lower-layer triggered mobility (LTM) must balance throughput against reliability under fast-varying radio conditions, where rare but costly events—handover and radio-link failures—dominate the user experience. We cast LTM HO as a Markov decision process and study distributional reinforcement learning (QR-DQN) with risk-aware action selection. We report two separable improvements over scalar DQN. First, the distributional learning target alone lowers the handover, ping-pong and failure rates at equal reward per decision. Second, a Conditional Value at Risk (CVaR) policy that acts on the lower tail of the return distribution drives the failure tails down further; the degree of risk aversion is a single monotone dial that trades a small amount of reward for reliability, with a knee at CVaR level $\alpha \approx 0.25$. We further show that the tail is best summarized by a single dedicated quantile: an agent that trains and acts on one quantile at $\tau = \alpha/2$ attains the lowest failure rates of any learned agent at the output size and inference cost of scalar DQN. On a 1,000-trajectory 5G LTM benchmark this risk-aware agent cuts the handover-failure rate of DQN by 40%, improves on a hand-tuned LTM baseline on every user-facing and safety KPI while issuing fewer handovers, and improves on a published contextual-bandit baseline in capacity, reliability and handover failures.

Index Terms—Distributional reinforcement learning, Quantile regression, Risk-sensitive control, 5G NR, Lower-layer triggered mobility, Handover optimization.

I. INTRODUCTION

Fifth-generation (5G) networks densify the radio access with many small, overlapping cells, so a moving user equipment (UE) crosses cell boundaries frequently and must be handed over from one cell—or sector—to another many times during a session. Each handover (HO) is a brief, risk-laden reconfiguration: chosen well it preserves throughput and reliability; chosen poorly it triggers a handover failure (HOF), a radio-link failure (RLF), or an unnecessary “ping-pong” (PP) back to the previous cell. *Lower-layer triggered mobility* (LTM) is the 5G New Radio mechanism that accelerates these decisions by triggering them from fast L1/L2 measurements rather than slower higher-layer signaling; the price of the extra speed is that the choice of *which* cell/sector to connect to at each measurement opportunity must be made under rapidly varying radio conditions. That selection is the decision problem we study.

Learning-based HO optimization typically casts cell selection as a heuristic rule or a contextual-bandit / scalar value-learning problem that maximizes an *expected* objective—throughput, or a weighted KPI score—sometimes augmented with reconfigurable intelligent surfaces (RIS) and LMMSE

channel prediction to improve the underlying signal quality [1]. Expected-value objectives are risk-neutral: they optimize the *average* outcome and are indifferent to its dispersion. Yet in LTM the user experience is dominated not by the average step but by rare, disproportionately costly tail events—handover and radio-link failures. The reward does penalize these events, so a mean-optimal policy is pushed to reduce their *average* frequency; but, optimizing only the expectation, it weighs each rare failure by its small probability and has little leverage to suppress the worst-case tail itself. This motivates a *risk-aware* formulation that acts on the lower tail of the outcome distribution rather than its mean.

In this paper we approach LTM HO with *distributional* reinforcement learning, which models the full return distribution $Z(x, a)$ —for a given radio/mobility state x (per-sector signal measurements, the serving cell and UE kinematics) and handover action a (stay, or switch to a prepared cell)—rather than its mean $Q(x, a)$, and with *risk-aware* action selection based on the Conditional Value at Risk (CVaR) of that distribution. Intuitively, learning the distribution yields a richer training signal, and acting on its lower tail makes the agent avoid worst-case outcomes instead of merely maximizing the average. Our contributions are:

- We formulate LTM HO as a Markov decision process and apply Quantile-Regression DQN (QR-DQN) with risk-aware action selection on a calibrated 5G LTM simulator (Sections III–IV).
- We show that the distributional learning target alone, acting on the mean, already improves the mobility and reliability KPIs over scalar DQN [10] at equal reward per decision (Section VI-A).
- We cast risk aversion as a single CVaR dial and place its three encodings—hard CVaR, spectral (soft-step) reweighting, and a single dedicated quantile—on a common reward–reliability frontier, finding a monotone trade with a knee near $\alpha = 0.25$ rather than an interior optimum (Section VI-B).
- We characterize *where to spend quantile capacity*: the mean is sharpened by adding quantiles, whereas the lower tail is best captured by a single dedicated quantile trained and acted on at $\tau = \alpha/2$ (“learn what you act on”), which attains the lowest failure rates of any learned agent at the output size and inference cost of scalar DQN (Section VI-C).

- On the 5G LTM benchmark we compare the risk-aware policy against a hand-tuned LTM baseline and the contextual multi-armed bandit (CMAB) of [1], improving on both across the user-facing and safety KPIs (Section VI-D).

II. BACKGROUND AND RELATED WORK

A. LTM and Handover Optimization

In legacy mobility the network reconfigures a UE’s serving cell through higher-layer (RRC) signaling driven by filtered L3 measurements—robust but slow. *Lower-layer triggered mobility* (LTM) instead prepares one or more candidate cells in advance and lets a fast L1/L2 measurement event trigger the execution of a pre-configured switch, cutting the interruption time but tightening the decision loop. The operational cost of a mobility policy is captured by a small set of events: a *handover* (HO) whenever the serving cell changes, a *ping-pong* (PP) when the UE returns to a recently left cell within a short window, a *handover failure* (HOF) when the target link cannot be established, and a *radio-link failure* (RLF) when the serving link collapses after a sustained run of out-of-sync observations within a time window. A good policy minimizes failures and ping-pongs while sustaining spectral efficiency. Prior learning-based work optimizes this trade-off with heuristics and contextual bandits over an expected-KPI objective, optionally stacking RIS and LMMSE channel prediction on top to improve signal quality and target pre-selection [1]; this is the closest prior art to ours. It is, however, risk-neutral—the gap we address.

B. Distributional Reinforcement Learning

Distributional RL predicts the full distribution of the return rather than its expectation. Instead of modeling the scalar action-value $Q(x, a)$ for each state x and action a , the agent models the return as a random variable $Z(x, a)$ that obeys the distributional Bellman equation:

$$Z(x, a) \stackrel{D}{=} R(x, a) + \gamma Z(X', A'), \quad (1)$$

where $\stackrel{D}{=}$ denotes equality in distribution, $R(x, a)$ is the reward, γ the discount factor, and (X', A') the (random) next state–action pair. Even when the policy ultimately acts on the mean $\mathbb{E}[Z(x, a)]$, learning the distribution provides a richer per-update signal and consistently improves the resulting policy. Most practical algorithms approximate the cumulative distribution as a staircase function; representative methods include C51 [2], QR-DQN [3], IQN [4] and FQF [5]. We build on QR-DQN, which parametrizes the distribution by the locations of a fixed set of quantiles.

C. Risk-Sensitive Reinforcement Learning

In standard distributional RL the greedy action maximizes $\mathbb{E}[Z(x, a)]$, approximated as the average of the N predicted quantiles $\frac{1}{N} \sum_{i=1}^N \theta_i(x, a)$. One may instead aggregate the θ_i under a different criterion to obtain a risk-sensitive policy. Distributions admit no canonical total order—under stochastic

dominance some pairs are simply incomparable—so any scalar aggregation imposes one, thereby encoding a particular risk attitude. A common risk-averse choice replaces the expectation with the Conditional Value at Risk at level α , CVaR_α , the average of the worst α -fraction of outcomes. Here α is a tail *fraction* that is averaged, not a quantile position: for a uniform return on $[-1, 1]$, $\text{CVaR}_{0.5}$ is the mean of the worse half of the outcomes—the values in $[-1, 0]$, i.e. -0.5 —whereas the median is the single $\tau=0.5$ quantile, 0. More generally, replacing the uniform average by a non-negative quantile weighting yields a *spectral* (or distortion) risk measure [6], with the expectation and CVaR_α as the uniform and lower-tail-uniform special cases; this weighted-quantile family is the design space for the risk-aware policies we study. In problems such as LTM HO, risk-aware decision-making is attractive because avoiding worst-case outcomes (handover or radio-link failures) is more valuable than maximizing the expected return alone. CVaR and related risk-sensitive criteria have a long history in reinforcement learning and operations research [7]–[9]; distributional RL makes them particularly convenient, since the return quantiles needed to evaluate CVaR_α are predicted directly by the distributional Q-value network.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. 5G LTM Deployment and Simulator

We evaluate on a calibrated multi-sector 5G deployment of 7 tri-sector base stations (21 sectors), driven by 1,000 pre-computed UE trajectories of 300 s each whose per-sector channel gains are sampled every 10 ms. The simulator advances at this resolution: at each tick it derives the serving signal-to-interference-plus-noise ratio (SINR) from the channel gains, transmit power and inter-cell interference, and maps it through a modulation-and-coding scheme (MCS) to a spectral efficiency; an SINR below the outage threshold yields zero rate, and a sustained run of outage ticks escalates to a radio-link failure (RLF). The radio and mobility constants (Table I) follow the reference LTM implementation of [1], and our LTM baseline reproduces that work’s published KPIs.

LTM maintains a small set of *prepared* candidate cells—at most 5, selected from the strongest neighbors by an RSRP threshold. In the standard heuristic a prepared cell is *executed* (the UE is handed over to it) once its RSRP exceeds the serving cell’s by a fixed margin. We keep this preparation machinery unchanged but replace the execution trigger with the learned agent: whenever at least one non-serving prepared cell exists, the agent decides *whether*—and to *which* prepared cell—to hand over. The decision cadence is therefore *event-driven*, the agent being consulted once per LTM preparation cycle, at intervals that vary with the radio dynamics. The agent controls only the *execution* decision—the part of LTM that trades reliability against signaling overhead—while preparation and the radio physics remain the reference LTM.

B. MDP Formulation

We model LTM HO as a Markov decision process in which the agent acts only at handover-decision opportunities. These

TABLE I
KEY SIMULATOR CONSTANTS (CALIBRATED TO THE REFERENCE LTM
IMPLEMENTATION).

Parameter	Value
Base stations $\times$ sectors	$7 \times 3 = 21$
UE trajectories	1,000 (300 s each)
Simulator step	10 ms
RL decision cadence	event-driven (per LTM cycle)
Transmit power	25 dBm
Noise floor	-101 dBm (-174 dBm/Hz $\times$ 20 MHz)
Receiver sensitivity	-95 dBm
Outage (RLF) SINR threshold	-3 dB
Preparation offset	-3 dB
Max. prepared cells	5
Execution offset (LTM trigger) [†]	3.0 dB

[†]The +3 dB execution offset is the LTM baseline's hard handover trigger; the RL agent replaces it with a learned, value-based decision.

arise once per LTM cycle and are unevenly spaced in time; we index the agent's successive decisions by t and treat each as one step.

State. An 88-dimensional vector: UE speed (1) and serving-cell tenure (1), the one-hot serving sector (21), per-sector RSRP (21), moving-average MCS (21) and moving-average SINR (21), and the UE position (x, y) (2).

Action. At each decision the agent chooses a target from the action set $\{\text{serving}\} \cup \{\text{prepared}\}$ —at most six entries, since preparation caps the candidate list at 5. Choosing the serving cell is a *stay* (no handover); choosing a non-serving prepared cell executes a handover to it. Because the agent itself fires the handover, it replaces LTM's +3 dB execution trigger with the learned value of acting now versus staying, while preparation and the candidate restriction remain the reference heuristic.

Reward. Each decision is rewarded over the interval until the next decision by the multiplicative LTM HO reward of [1],

$$R_t = R_t^{\text{thr}} \cdot \frac{\alpha_{\text{HO}}^{\text{HO}} \alpha_{\text{PP}}^{\text{PP}} \alpha_{\text{HOF}}^{\text{HOF}}}{1 + e^{2(N_t^{\text{OOS}} - 2)}}, \quad (2)$$

where R_t^{thr} is the throughput term—the serving spectral efficiency (bps/Hz) *averaged over the decision interval*; the indicators $\mathbb{I}_t^{\text{HO}}, \mathbb{I}_t^{\text{PP}}, \mathbb{I}_t^{\text{HOF}} \in \{0, 1\}$ flag whether a handover, ping-pong or handover failure occurred *anywhere within* that interval; and $\alpha_{\text{HO}}, \alpha_{\text{PP}}, \alpha_{\text{HOF}} \in (0, 1]$ are the corresponding multiplicative penalties (defaults $\alpha_{\text{HOF}}=0.1$, $\alpha_{\text{HO}}=0.8$, $\alpha_{\text{PP}}=0.9$, so a failure is penalized far more harshly than a routine handover). The final factor is a reverse sigmoid of the out-of-sync counter N_t^{OOS} (the number of consecutive out-of-sync indications on the serving link): it stays near 1 while the link is in sync, equals 0.5 at $N_t^{\text{OOS}}=2$, and decays toward 0 as the link approaches radio-link failure. Since the penalty factors lie in $(0, 1]$, the reward is the portion of the attainable throughput R_t^{thr} that the policy retains after mobility events and link instability, which the distributional agent models as a random return. Because decisions are event-driven, a policy's own choices change how many actions it faces per episode; we therefore compare policies by their mean reward *per decision* rather than the episode sum, which would otherwise credit a policy merely for being consulted more often (Section V).

IV. METHOD

A. QR-DQN and the Quantile Loss

QR-DQN parametrizes $Z(x, a)$ by its inverse CDF at N quantile fractions $\tau_1, \dots, \tau_N \in [0, 1]$, learning $\theta_i(x, a) \approx F_{Z(x, a)}^{-1}(\tau_i)$. These $\theta_i(x, a)$ are the outputs of a neural network: a shared trunk maps the state x to an output layer (the *head*) that emits the N quantile estimates for every action a , and the network parameters ϕ are fit by minimizing the loss defined below. Vanilla QR-DQN places these at the midpoints $\tau_i = (i - \frac{1}{2})/N$ with uniform weights $w_i = 1/N$. The aggregated value used for action selection,

$$\mathbb{E}[Z(x, a)] = \int_0^1 F_{Z(x, a)}^{-1}(\tau) d\tau \approx \sum_{i=1}^N w_i \theta_i(x, a), \quad (3)$$

is a finite-sample quadrature with weights $w_i \geq 0$, $\sum_i w_i = 1$; the pair (τ_i, w_i) defines the *quantile scheme*, and the midpoint choice is the midpoint Riemann sum for $\int_0^1 F^{-1}(\tau) d\tau$. Risk sensitivity is introduced later (Section IV-B) by altering this aggregation and, for some variants, the placement of the trained quantiles; the loss below keeps the same form throughout.

The quantiles are trained with the quantile (pinball) Huber loss. Writing the Huber function with threshold κ as

$$\mathcal{H}_\kappa(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \kappa, \\ \kappa(|u| - \frac{1}{2}\kappa), & |u| > \kappa, \end{cases} \quad (4)$$

the quantile-Huber penalty at fraction τ is

$$\rho_\tau^\kappa(u) = |\tau - \mathbb{I}\{u < 0\}| \mathcal{H}_\kappa(u), \quad (5)$$

The Huber threshold κ rounds the kink of the plain pinball (absolute-value) loss into a quadratic region near zero, giving bounded, continuous gradients that are robust to the large bootstrapped errors common early in training [3]. The asymmetric prefactor $|\tau - \mathbb{I}\{u < 0\}|$ is what makes this *quantile* regression rather than ordinary regression: it penalizes errors asymmetrically—weight τ on one side of zero and $1 - \tau$ on the other—so the per-quantile loss is minimized when θ_i sits at the τ_i -quantile of its target rather than at the mean. The per-transition loss sums the pairwise temporal-difference errors between each of the N *predicted* quantiles (the online network's current outputs) and each of the N *target* quantiles $y_j = r + \gamma \theta_j(x', a^*)$ (bootstrapped from a frozen target network),

$$\mathcal{L}(x, a) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \rho_{\tau_i}^\kappa(y_j - \theta_i(x, a)), \quad (6)$$

with a^* the greedy action under the aggregated value (3). Equation (6) is best read as a sum of independent per-quantile regressions. Ideally each prediction $\theta_i(x, a)$ would be fit to its *expected* quantile-Huber loss against the bootstrapped target $Y = r + \gamma Z(x', a^*)$ (i.e. minimizing $\mathbb{E}[\rho_{\tau_i}^\kappa(Y - \theta_i(x, a))]$),

whose minimizer is the τ_i -quantile of Y). Since Y is unavailable in closed form, we represent it by its own N predicted quantiles $\{y_j\}$ and average their losses (the $\frac{1}{N}$ factor). Every predicted quantile is therefore regressed against *all* N targets—the whole bootstrapped return distribution—rather than a single scalar value, and it is this distribution-to-distribution regression that drives the network to recover the shape of $Z(x, a)$.

B. Risk-Aware Quantile Aggregation

The quantile loss (6) fixes how the return distribution is learned but leaves the aggregation (3) free. A risk-sensitive policy keeps the same network and replaces the uniform average by a non-negative weighting

$$Q_\omega(x, a) = \sum_{i=1}^N \omega_i \theta_i(x, a), \quad \omega_i \geq 0, \sum_i \omega_i = 1, \quad (7)$$

acting greedily on Q_ω (so ω also selects a^* in (6)). When ω_i is non-increasing in τ_i this is a *spectral* (distortion) risk measure (Section II-C), with the expectation as its uniform special case. We make the policy tail-averse in two ways: by *re-weighting* the quantiles of a faithfully trained distribution (the hard-CVaR and soft-step policies below, with the mean as the risk-neutral baseline), which leaves (6) untouched and therefore adds nothing at inference; or by *re-placing* the trained quantiles onto the tail (truncation and the single-quantile policy), which instead concentrates network capacity where the policy acts.

Risk-neutral (mean). Setting $\omega_i = w_i = 1/N$ recovers the expectation; this is our risk-neutral (RN) agent, the direct distributional counterpart of DQN. It carries no tail preference but still benefits from the richer per-update signal of (6).

Hard CVaR. The standard risk-averse criterion is the conditional value at risk at level α , the mean of the worst α -fraction of returns—on the midpoint grid, a uniform average over the lowest $k = \lceil \alpha N \rceil$ of the N quantiles (ordered by increasing τ_i):

$$\omega_i^{\text{CVaR}} = \frac{\mathbb{I}\{i \leq k\}}{k}, \quad (8)$$

a hard 0/1 step at the k -th quantile. The upper $N - k$ quantiles are trained but never consulted at decision time.

Soft-step (spectral). Hard CVaR is the limit of a one-parameter family that relaxes the step to two finite levels: the lowest k quantiles receive $r \geq 1$ times the weight of the rest,

$$\omega_i(r) = \frac{r \mathbb{I}\{i \leq k\} + \mathbb{I}\{i > k\}}{r k + (N - k)}. \quad (9)$$

Here $r=1$ gives back the mean (RN) and $r \rightarrow \infty$ recovers hard CVaR $_\alpha$ (8), with intermediate r interpolating a graded tail emphasis. The positions stay uniform—all N quantiles remain active and faithfully trained—so risk lives entirely in the weights, and every member is a valid spectral measure. This realizes the weighted-quantile design space anticipated in Section II-C.

Truncated CVaR. Equation (8) wastes capacity: the head predicts N quantiles but only the lowest k enter the decision. The truncated variant trains *only* k quantiles, placed uniformly

in $[0, \alpha]$; their plain mean equals CVaR $_\alpha$ exactly under the midpoint rule. This realizes the same risk policy with a smaller head and a tail-only loss. Concentrating capacity on the consulted region might be expected to sharpen the estimate; Section VI-C shows instead that the full and truncated CVaR policies are statistically indistinguishable on reward per decision and tail KPIs—truncation buys lower cost and faster training, not a better policy.

Single dedicated quantile (RA-1q). Taken to its extreme, truncation keeps a single quantile ($k=1$) at $\tau = \alpha/2$ —the center of the tail lobe—and selects on it directly, a VaR-style point criterion. RA-1q learns and acts on exactly one return statistic, matching the output dimension and inference cost of scalar DQN. The link is sharpest at the risk-neutral end: with $N=1$, $\tau=0.5$ the asymmetric weight $|\tau - \mathbb{I}\{u < 0\}|$ in (6) becomes constant ($\frac{1}{2}$) and the loss reduces to a symmetric Huber regression onto a single robust value (the median in the un-Huberized limit)—the distributional counterpart of plain value learning. DQN and RA-1q are therefore two ends of a single-quantile family separated only by the acted fraction τ (0.5 vs. $\alpha/2$). This “learn what you act on” design is tested in Section VI-C against both spreading capacity over a tail lobe (truncated and full CVaR) and its converse—learning the full distribution but acting on only its $\tau=\alpha/2$ quantile. Section VI-B places these knobs—hard CVaR, soft-step weights, and the single dedicated quantile—on a common risk-reliability frontier.

V. EXPERIMENTAL SETUP

A. Evaluation Protocol and Metrics

All agents are trained and evaluated on the same 1,000 pre-computed UE trajectories (300 s each, 10 ms simulation resolution), with the agent consulted at the event-driven handover-decision opportunities of Section III-B. We deliberately use the full trajectory set for both training and final evaluation rather than a held-out split: the channel-gain dataset is fixed and the policy never observes future returns, so the relevant question is how well each agent exploits the available propagation environment, not generalization to unseen geometry. Each final model is trained for 2,000 episodes across 5 random seeds; the comparative studies in Section VI use a lighter 1,000-episode, 3-seed setting. After training, we evaluate every agent with frozen weights and greedy action selection ($\epsilon=0$), replaying all 1,000 UE trajectories, and report the cross-seed mean $\pm$ standard deviation throughout. The full implementation (simulator, agents and evaluation scripts) is publicly available [11].

Following Section III-B, we report the mean *reward per decision*—the episode reward divided by the number of decisions taken—as the primary scalar objective, since the episode sum otherwise grows simply with the number of decisions a policy triggers. Alongside it we track the eight physical KPIs of [1]: handover rate (HO/min), handover-failure rate (HOF/min), ping-pong rate (PP/min), radio-link-failure rate (RLF/min), spectral efficiency / capacity (bps/Hz), link reliability (%), cell preparations (/min) and resource reservation (%).

B. Models and Hyperparameters

We compare four learned agents that share an identical two-layer MLP trunk (256×256) and the same observation, action and reward definitions, so that differences trace to the *learning target* and the *risk rule* rather than to trunk capacity or input. They differ only in the output head and how it is read for action selection (Section IV-B):

- 1) **DQN**—scalar value learning; greedy $\arg \max_a Q(x, a)$.
- 2) **RN**—risk-neutral QR-DQN; an $N=25$ midpoint head read by the mean policy $\mathbb{E}[Z(x, a)]$.
- 3) **Soft-step**—the *same* $N=25$ head, read instead through the soft-step spectral weighting (9) at $\alpha=0.25$ and ratio $r=2$ (all 25 quantiles re-weighted, none discarded).
- 4) **RA-1q**—a single dedicated quantile ($k=1$) at $\tau=\alpha/2$ ($\alpha=0.25$), matching the scalar head size and inference cost of DQN.

These are benchmarked against two references: our own hand-tuned LTM baseline and the contextual-bandit (CMAB) policy of [1]. We additionally train full- and truncated-CVaR agents, soft-step agents at several ratios r , and single-quantile agents (Section VI), each differing only in the aggregation or trained grid of Section IV-B.

All agents bootstrap from a frozen *target network* whose parameters ϕ^- track the online parameters ϕ by a soft update, $\phi^- \leftarrow (1-\tau)\phi^- + \tau\phi$ with $\tau=0.01$, rather than a periodic hard copy; the greedy next action and its quantiles are both read from this target network. Soft and hard updates are within cross-seed noise here, so we fix the soft scheme for all agents. The remaining hyperparameters (Table II) follow coarse grid searches and the prior work [1], with one gradient update every eight environment steps for training speed; network width saturates at 256×256 , and n -step returns, gradient clipping and optimizer- ϵ variants gave no improvement. The Huber threshold $\kappa=0.25$ was selected from a coarse sweep over $\{0.25, 0.5, 1, 2\}$; smaller κ gave marginally higher reward and lower failure rates, and we fix $\kappa=0.25$ throughout.

TABLE II

SHARED TRAINING HYPERPARAMETERS. THE QUANTILE COUNT APPLIES TO RN AND SOFT-STEP; RA-1Q USES A SINGLE QUANTILE ($k=1$), AND κ APPLIES TO ALL THREE QR-DQN AGENTS. DQN IS A SCALAR VALUE LEARNER; EACH AGENT’S RISK RULE (MEAN, SOFT-STEP, OR SINGLE QUANTILE) IS GIVEN IN THE AGENT LIST OF SECTION V-B.

Hyperparameter	Value
Hidden layers	256×256
Optimizer	Adam
Learning rate	10^{-4}
Discount γ	0.9
Replay buffer	10^5
Batch size	256
Train frequency	8
Target update	soft, $\tau=0.01$
ϵ schedule	$1.0 \rightarrow 0.05$ (geom., $\times 0.995/\text{ep}$)
Quantiles N (RN, soft-step)	25, midpoint
Huber κ (QR-DQN)	0.25

VI. RESULTS ON THE LTM HANDOVER TASK

A. Distributional Learning on the Handover Task

We first establish how the two learning agents behave on the handover task before turning to risk. Scalar DQN and risk-neutral QR-DQN (RN) are trained on the full set of 1,000 trajectories for 2,000 episodes and evaluated, with results averaged over five seeds. The two differ only in the output layer: DQN regresses a single scalar value, whereas RN predicts a full distributional head (Section IV), here 25 quantiles, and acts on its mean. Table III reports both, alongside a single-quantile RN that isolates the effect of the distributional representation.

Distributional learning versus the scalar baseline. DQN and RN earn the same reward per decision, yet RN improves on every other axis: it lowers the handover, ping-pong, and handover-failure rates, raises capacity and link reliability, and leaves resource use essentially unchanged (Table III). For scale, a uniform-random execution policy—choosing uniformly among the serving and prepared cells, evaluated under the same protocol—earns only 1.93 reward per decision and the LTM heuristic 3.544 (Section VI-D); both learned agents therefore sit near the achievable ceiling and separate on the mobility and reliability KPIs rather than on reward itself. Learning the return distribution and acting on its mean improves the scalar baseline at equal reward.

Quantile count for the mean. The improvement comes from the full distribution, not from the quantile loss alone. A risk-neutral head reduced to a single quantile at $\tau = 0.5$ closely reproduces DQN across every metric (Table III, middle column): as Section IV-B notes, a lone median quantile trained under the Huber loss recovers essentially the same robust value that DQN learns on these near-symmetric returns. The reliability gain grows as further quantiles populate the head, with the largest step in the first few and diminishing returns afterward: sweeping N at the lighter study setting (1,000 episodes, three seeds), the handover-failure rate falls from 0.64/min at $N=1$ to 0.55/min by $N=3$ and eases further to 0.50/min by $N=25$ (handover rate 6.0, 5.3, 5.0/min), while the reward per decision stays flat (≈ 3.66). We therefore carry the $N=25$ head throughout; at the final 2,000-episode, five-seed setting it remains the best-performing mean policy (Table III, right column), and it supplies the finer tail grid that the CVaR and soft-step risk studies act on ($k = \lceil \alpha N \rceil$ tail quantiles, Section VI-B). The quantiles’ *placement*, by contrast, is immaterial: replacing the midpoint grid with Gauss–Legendre, trapezoidal, Simpson, or Beta-warped positions leaves the reward per decision within cross-seed noise (between 3.654 and 3.663), so we keep the simple midpoint grid throughout.

B. A Controllable Risk Dial

Having shown that the distributional representation helps, we now make the policy risk-averse and ask how the degree of risk-aversion trades against performance. The CVaR level α is a single dial: $\alpha = 1$ recovers the risk-neutral mean, and smaller α focuses the policy on the lower tail of the return

TABLE III

DISTRIBUTIONAL LEARNING (2,000 EPISODES, FIVE SEEDS). THE THREE COLUMNS ISOLATE THE SOURCE OF THE GAIN: SCALAR DQN; A DISTRIBUTIONAL HEAD COLLAPSED TO ONE MEDIAN QUANTILE ($N=1$), WHICH REPRODUCES DQN; AND THE FULL 25-QUANTILE HEAD—THE RN AGENT USED THROUGHOUT THE PAPER—WHICH LOWERS THE HANDOVER, PING-PONG, AND FAILURE RATES AT EQUAL REWARD. BOLD MARKS THE BEST VALUE IN EACH ROW.

KPI	DQN	RN ($N=1$)	RN ($N=25$)
Reward/decision $\uparrow$	3.667 ± 0.006	3.669 ± 0.003	3.663 ± 0.011
Capacity (bps/Hz) $\uparrow$	4.062	4.072	4.068
Reliability (%) $\uparrow$	97.00	97.02	97.07
HO/min $\downarrow$	5.63	5.63	5.12
PP/min $\downarrow$	1.065	1.079	0.857
HOF/min $\downarrow$	0.667	0.648	0.551
RLF/min $\downarrow$	0.047	0.048	0.046
Prep/min $\downarrow$	28.1	27.6	27.7
Res. reserv. (%) $\downarrow$	8.85	8.74	8.91

distribution. We sweep it over 1,000 episodes and three seeds on the $N=25$ midpoint head (hard CVaR averaging the lowest $k = \lceil \alpha N \rceil$ quantiles), and defer how the tail estimate should be built to Section VI-C.

Risk trades monotonically against reward. Table IV dials α from mild (0.5) to aggressive (0.1). As α decreases the policy grows more conservative: it hands over less and fails less, while giving up a little reward, capacity, and link reliability. The trade is monotone and has no interior optimum—there is no single “best” risk level, only a controllable exchange of reward for reliability. The reward cost is mild while the failure reduction levels off, leaving a clear knee near $\alpha = 0.25$ (also visible in the frontier of Fig. 1), which we adopt.

The risk–reliability frontier. The three risk encodings of Section IV-B—averaging the lower tail (hard CVaR), reweighting a faithfully trained grid (soft-step), and committing to one dedicated quantile—can all be placed on a single risk–reliability frontier (Fig. 1). Soft-step is its continuous member (9): at ratio $r = 1$ it is the risk-neutral mean, and as r grows it shifts weight onto the tail and approaches hard CVaR, the reward per decision moving from 3.655 at $r = 2$ to 3.606 at $r = 100$, toward the hard-CVaR value of 3.598. We adopt $r = 2$ for the headline Soft-step agent: it captures most of the reliability gain at the favorable, low-curvature end of the frontier, whereas larger r pays increasing reward for diminishing returns. The single dedicated quantile lies on the most favorable edge of the frontier, delivering more reward at any failure level than the averaging or reweighting schemes; we examine why in Section VI-C.

C. Where to Spend Quantile Capacity

Section VI-A showed that the *mean* is sharpened by adding more quantiles, with diminishing returns. The risk estimate behaves oppositely: the lower tail is best summarized by very few quantiles, and in the limit by a single dedicated one. The studies below use 1,000 episodes and three seeds; the headline single-quantile agent is also reported at the final 2,000-episode, five-seed setting.

Quantile count for the tail. Holding the risk level at $\alpha = 0.25$ and averaging the lowest k quantiles (ordered by τ) into

TABLE IV

THE HARD-CVaR RISK DIAL ON THE $N=25$ HEAD (1,000 EPISODES, THREE SEEDS), FROM MILD ($\alpha = 0.5$) TO AGGRESSIVE ($\alpha = 0.1$). GREATER RISK-AVERSION LOWERS THE HANDOVER AND FAILURE RATES AT A MODEST COST IN REWARD, CAPACITY, AND LINK RELIABILITY; THE TRADE IS MONOTONE, WITH A KNEE NEAR $\alpha = 0.25$. BOLD MARKS THE BEST VALUE IN EACH ROW (TIES AT THE SHOWN PRECISION BROKEN BY THE UNROUNDED VALUE).

KPI	$\alpha=0.5$	$\alpha=0.25$	$\alpha=0.1$
Reward/decision $\uparrow$	3.613 ± 0.016	3.598 ± 0.003	3.542 ± 0.014
Capacity (bps/Hz) $\uparrow$	4.048	4.019	3.974
Reliability (%) $\uparrow$	97.00	96.90	96.71
HO/min $\downarrow$	4.76	4.33	4.17
PP/min $\downarrow$	0.826	0.627	0.498
HOF/min $\downarrow$	0.442	0.367	0.353
RLF/min $\downarrow$	0.045	0.043	0.043
Prep/min $\downarrow$	27.6	26.0	25.9
Res. reserv. (%) $\downarrow$	9.3	9.3	9.4

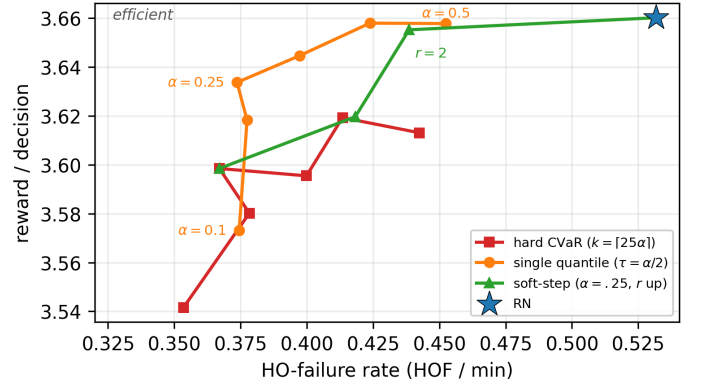


Fig. 1. Risk–reliability frontier (1,000 episodes, three seeds): reward per decision against the handover-failure rate for the three risk encodings, with the risk-neutral agent (RN) as the least risk-averse anchor. The single dedicated quantile (orange) is the efficient frontier—more reward at any failure level—with its knee at $\alpha = 0.25$; soft-step interpolates continuously between RN and hard CVaR. Upper-left is better.

the CVaR estimate, the reward per decision falls steadily as k grows—from 3.634 at $k = 1$ to 3.555 at $k = 25$ —while the failure rate barely moves. Packing the tail more densely does not help either: at $\alpha = 0.4$ a densely truncated head (3.587) is worse than a plain one (3.619). Where the mean was sharpened by more quantiles, the tail estimate is only diluted by them.

The single dedicated quantile. The limit $k = 1$ —one quantile trained and acted on at $\tau = \alpha/2$ (RA-1q)—is the most efficient point of the study. At the final setting it attains the lowest failure rates of any learned agent (HOF 0.399/min, against DQN’s 0.667; Table V) while matching the output size and inference cost of scalar DQN, and it trains fastest of all (76 min, versus 87 for DQN and 141 for the 25-quantile RN). Committing the network’s one statistic to the quantile the policy actually consults—“learn what you act on”—is both the cheapest and the most failure-averse design.

Training the acted quantile. Acting on a single tail quantile is not enough on its own; that quantile must also be the one trained. The converse design—learning the full 25-quantile distribution but selecting on a single $\tau = \alpha/2$ quantile—ties RA-1q on reward yet leaves the failure rate markedly

higher (HOF 0.42–0.48 versus RA-1q’s 0.374 in the 1,000-episode studies), while fitting 25 quantiles instead of one costs substantially more to train. Concentrating the loss on the single acted quantile is what sharpens the tail.

Full versus truncated heads. The CVaR policy itself can be realized two ways: a full $N = 25$ head whose lowest $k = \lceil \alpha N \rceil$ quantiles are averaged, or a truncated head that trains only those k over $[0, \alpha]$. The two are statistically indistinguishable on reward per decision and tail KPIs (at $\alpha = 0.4$, 3.628 versus 3.619); truncation merely trains faster with a smaller head (65 min versus 79 min). It is an efficiency option, not a better estimator.

D. Comparison with the LTM Heuristic and Contextual Bandits

Table V and Fig. 2 report the four learned agents at the final 2,000-episode, five-seed setting against two references: our hand-tuned LTM baseline, run in the same simulator and therefore directly comparable, and the contextual multi-armed bandit (CMAB) of [1], whose published figures come from a different simulator and are indicative only (Table V, footnote).

Versus the LTM heuristic. Every learned agent improves on the heuristic across the user-facing and safety KPIs, and the risk-aware RA-1q does so by the widest margin: relative to LTM it raises capacity by 7.8% (4.044 versus 3.753 bps/Hz) and link reliability by 1.7 points (96.94% versus 95.25%), and it cuts the handover-failure rate by 65% (0.399 versus 1.151/min), the ping-pong rate by 82% (0.656 versus 3.625), and the radio-link-failure rate by 31% (0.045 versus 0.065). It also earns more reward per decision than the heuristic (3.641 versus 3.544), so the advantage holds on the training objective itself.

Signaling frugality. The learned policies are also less active than either reference, not more. In the event-driven setting they issue far fewer handovers—4.6 to 5.6 per minute, against 11.3 for LTM and 8.0 for CMAB—and correspondingly fewer ping-pongs. The distributional agents thus achieve their reliability and throughput gains while *reducing* control-plane mobility, not by trading signaling for reliability.

The resource-reservation trade-off. The one metric on which the LTM baseline leads is resource reservation: the learned agents hold more cells reserved (8.8 to 9.0%, versus 6.4% for LTM and 6.8% for CMAB). Keeping more candidates ready is the cost they pay for their lower failure rates; preparation rates, by contrast, are comparable across all agents (24 to 28/min, lowest for CMAB at 24.1 and RA-1q at 24.7).

Relation to contextual bandits. The CMAB of [1] addresses the *same* decision as our agent: at each opportunity it selects the handover target among the candidate cells from contextual features, trained on the *same* multiplicative reward (2). It is therefore a direct alternative to our policy, not a complementary mechanism. The two differ in *how* the decision is learned: CMAB is a contextual bandit that treats each decision in isolation—assuming the chosen action does not affect future context—and optimizes the expected reward, whereas our agent is a sequential, bootstrapped value learner

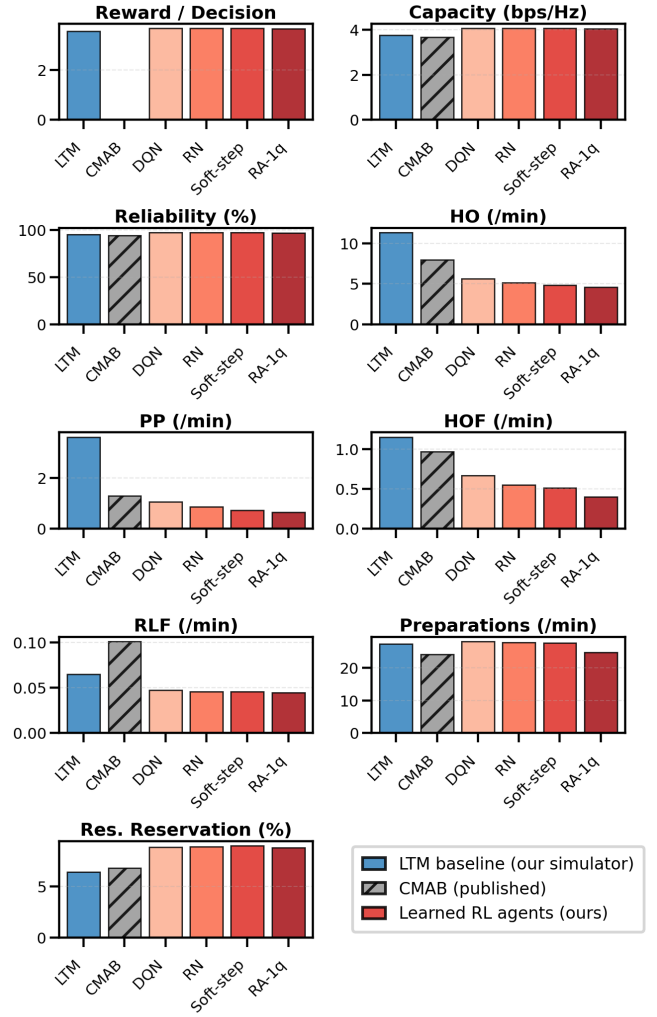


Fig. 2. Final KPI comparison on the LTM handover task (exact values in Table V). References first—our LTM baseline (blue) and the published CMAB (grey, hatched, from [1])—then the four learned agents (red, lightest to darkest: DQN, RN, Soft-step, RA-1q). All panels start at zero; ↓ KPIs (HO, PP, HOF, RLF, preparations, resource reservation) are better when lower. Reward per decision is our training objective; CMAB reports none.

that additionally acts on the lower tail of the return distribution. On the user-facing quality KPIs the risk-aware agent improves on the reported CMAB figures (capacity 4.044 versus 3.68, reliability 96.94% versus 94.0%, handover-failure rate 0.399 versus 0.97), subject to the cross-simulator caveat above.

VII. DISCUSSION, LIMITATIONS, AND FUTURE WORK

Limitations. Several caveats bound our claims. (i) Training and evaluation use the *same* fixed 1,000-trajectory set with no held-out split; this is the appropriate protocol for the question we ask—how well a policy exploits a fixed propagation environment, matching the evaluation of [1]—but it does not measure generalization to unseen geometry. (ii) We study a single deployment and channel-gain dataset. (iii) The reward-shaping coefficients are inherited from [1] and not re-tuned here; the Huber threshold κ , the network width and the

TABLE V

FINAL PERFORMANCE ON THE LTM HANDOVER TASK. FOR THE LEARNED AGENTS, THE REWARD PER DECISION IS THE CROSS-SEED MEAN $\pm$ STANDARD DEVIATION OVER 2,000 EPISODES AND FIVE SEEDS (STANDARD DEVIATION ACROSS SEEDS, NOT A CONFIDENCE INTERVAL; EACH SEED IS EVALUATED ON ALL 1,000 TRAJECTORIES). **LTM**: OUR HAND-TUNED BASELINE. **CMAB***: THE CONTEXTUAL MULTI-ARMED BANDIT OF [1]. **DQN**: SCALAR BASELINE. **RN**: RISK-NEUTRAL QR-DQN ($N=25$). **SOFT-STEP**: SPECTRAL REWEIGHTING ($\alpha=0.25$, $r=2$). **RA-1q**: SINGLE DEDICATED QUANTILE ($\alpha=0.25$). THE LTM REWARD PER DECISION IS MEASURED IN OUR SIMULATOR; CMAB REPORTS NO REWARD. BOLD MARKS THE BEST VALUE IN EACH ROW ACROSS ALL AGENTS; $\uparrow/\downarrow$ GIVE THE DESIRABLE DIRECTION.

KPI	LTM	CMAB*	DQN	RN	Soft-step	RA-1q
Reward/decision $\uparrow$	3.544	—	3.667$\pm$0.006	3.663 $\pm$ 0.011	3.659 $\pm$ 0.007	3.641 $\pm$ 0.008
Capacity (bps/Hz) $\uparrow$	3.753	3.68	4.062	4.068	4.067	4.044
Reliability (%) $\uparrow$	95.25	94.0	97.00	97.07	97.08	96.94
HO/min $\downarrow$	11.34	8.00	5.63	5.12	4.85	4.59
PP/min $\downarrow$	3.625	1.30	1.065	0.857	0.732	0.656
HOF/min $\downarrow$	1.151	0.97	0.667	0.551	0.512	0.399
RLF/min $\downarrow$	0.065	0.101	0.047	0.046	0.046	0.045
Prep/min $\downarrow$	27.3	24.1	28.1	27.7	27.6	24.7
Res. reserv. (%) $\downarrow$	6.44	6.82	8.85	8.91	9.00	8.82

*CMAB figures are reproduced from [1] and were obtained in that work’s simulator; our LTM baseline reproduces that work’s published LTM KPIs to within $\approx 5\%$ on the radio and mobility metrics, so the CMAB column is indicative rather than strictly comparable.

quantile placement, by contrast, are swept (Sections V, VI-A) and have little effect on the operating point. (iv) We explore CVaR and its spectral (soft-step) relaxations, but not the wider space of coherent or distortion risk measures, which may trade the KPIs off differently.

Future work. Several directions follow naturally. Implicit and fully-parameterized quantile networks (IQN [4], FQF [5]) learn the quantile fractions themselves and would subsume the quantile-placement study. Constrained or multi-objective formulations could bound the handover- or radio-link-failure rate directly, rather than through the scalar reward. Our agent also learns only the *execution* decision while the candidate *preparation* stage stays fixed to the LTM heuristic; learning a risk-aware preparation policy—not only execution—is a natural extension, as is closing the one gap where the learned policies trail the heuristics, their higher resource reservation, which a reservation-aware reward term could price in directly. Finally, it remains to confirm that the distributional and quantile-placement findings transfer to standard finite-action RL benchmarks such as the Atari 2600 suite.

VIII. CONCLUSION

On the LTM handover task, distributional RL delivers two separable improvements over scalar DQN. First, the distributional learning target alone—the risk-neutral RN agent—lowers the handover, ping-pong and failure rates and raises capacity and reliability at equal reward per decision. Second, making the policy risk-averse with a CVaR criterion drives the failure tails down further, trading a small, controllable amount of reward for reliability along a monotone risk dial with a knee near $\alpha = 0.25$. The most effective realization is the single, dedicated quantile (RA-1q): trained and acting on one tail quantile, it attains the lowest handover, ping-pong and failure rates of any learned agent—cutting DQN’s handover-failure rate by 40%—at the output size and inference cost of scalar DQN. Against our hand-tuned LTM baseline in the same simulator it improves every user-facing and safety KPI while issuing *fewer* handovers, and it improves on the published contextual-bandit figures on the quality KPIs; the only metric

on which the LTM baseline leads is resource reservation. For the lower tail that governs handover reliability, one well-placed quantile is enough—“learn what you act on.”

ACKNOWLEDGMENTS

The author thanks Josep Vidal and Marga Cabrera for their mentoring and guidance throughout this work, and Ainna Yue Moreno-Locubiche for providing the base LTM simulation code and for help in debugging the simulator.

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